7.1 → Bergstrat-Sherman techniques

→ Bergstrat-Sherman claims settlement rate adjustment

→ Why necessary? → A change in the settlement rates can distort the age-to-age factors using the development method

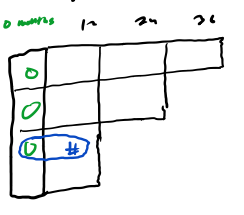
→ ① Check if needed

→ Calculate disposal rates ▽, if significant changes over ⇒ then continue

→ ② Select disposal rates

→ ③ Adjust cumulative paid claims ⇒ (a) restate ▽

Original Cumulative paid Claims



→ ④ Perform the paid development method like usual on ② adj cumulative paid claims

⇒ estimated ultimate claims

→ Bergstrat-Sherman case outstanding adequacy adjustment

→ adjusts losses for change in case reserve adequacy

⇒ If strengthening case reserves, using regular dev method will overstate results b/c LDFs calculated on the unadjusted data would be applied to higher rept loss (from the strengthened reserves) will overstate the ultimate b/c

→ Shortcoming in it is based highly on the selected severity

↳ think about any assumptions made in the calculations

→ ⓪ To determine if necessary, check if avg case outstanding trends the same as severity

↳ = case outstandings → = rept claims − paid claims / open claims

→ ① Trend the most recent avg case outstanding backwards in time

▽ (using the severity trend, b/c they should align)

→ ② Calculate adjusted cumulative rept claims = adj avg case outstandings * open claim count + paid claims

▽

→ ③ Then do rept dev method like usual

⇒ estimated ultimate claims

Accident Year	Adjusted Average Case Outstanding ($000)			
	12 Months	24 Months	36 Months	48 Months
2011	$5.922(1.05)^{-3} = 5.116$	$5.060(1.05)^{-2} = 4.590$	$4.714(1.05)^{-1} = 4.490$	0
2012	$5.922(1.05)^{-2} = 5.372$	$5.060(1.05)^{-1} = 4.819$	4.714	
2013	$5.922(1.05)^{-1} = 5.640$	5.060		
2014	5.922			

8.1 → Reinsurance: Salvage + Subrogation

→ Salvage + Subrogation

→ Final formula

unpaid SOS = ultimate SOS − Received SOS

or using reported SOS:

unreported SOS = ultimate SOS − Reported SOS

"recoverable" → recoverable SOS means unpaid SOS ⇒ If unsure, give both answers w/ an explanation

→ Methods

→ #1) Development method

⇒ Perform the usual directly on SOS dollars

→ #2) SOS ratio method → ① calculate SOS ratios ▽▽

② Calculate LDFs & select

③ Calculate CDFs

④ Develop SOS ratios → select w/ ratio (use simple options, just justify)

⑤ use ④ & % by rept years gross of SOS

⑥ Annualize to get final amount of interest

→ Advantages → i) can use actuarial judgement to select w/ SOS ratio

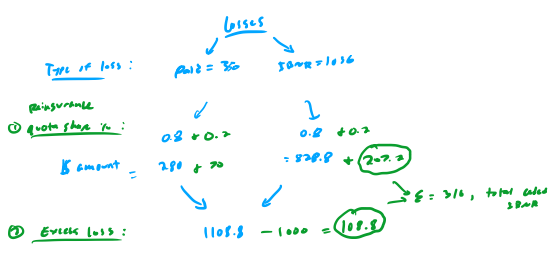
ii) more stable than dev methods

⇒ iii) LDFs using this approach are less leveraged than LDFs of SOS $

→ Reinsurance

→ DRAW out reinsurance contracts ⇒ Do one-at-a-time & in pieces

↳ paid + unpaid (IBNR)



000 (running out of time)

→ Kittle refinement for estimating unpaid ULAE

→ ① calculate w_i = Paid ULAE$_i$ / 0.5(Paid claims$_i$ + Incurred claims$_i$)

→ ② Select w^*

→ ③ Calculate unpaid ULAE = w^*[Pure IBNR + 0.5(case res + IBNER)]

↳ = 0 for claims-made policies

↳ Development of known claims